



A Framework for Phoneme-Level Pronunciation Assessment Using CTC

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Abstract

Traditional phoneme-level goodness of pronunciation (GOP) methods require phoneme to speech alignment. The drawback is that these methods, by their definitions, are prone to alignment errors and preclude the possibility of deletion and insertion errors in pronunciation. We produce experimental evidence that CTC-based methods can be used in traditional GOP estimation in spite of their “peaky” output behaviour and may be less prone to alignment errors than traditional methods. We also propose a new framework for GOP estimation based on CTC-trained model that is independent of speech-phoneme alignment. By accounting for deletion and insertions as well as substitution errors, we show that our framework outperform alignment-based method. Our experimental results are based on the CMU-kids dataset for child speech and on the Speechocean762 containing both child and adult speech speakers. Our best method achieves 29.02% relative improvement over the baseline GOP methods.

Index Terms: pronunciation assessment, goodness of pronunciation, end-to-end, CTC, child speech¹.

1. Introduction

Computer-Aided Language Learning (CALL) and Computer-assisted pronunciation training (CAPT) are drawing more attention from language learners and teachers in recent years due to their ubiquitous availability and high degree of self-controlled manner of study. Mispronunciation detection and/or diagnosis (MDD), which aims at detecting and correcting mispronunciations of the learners on interaction, is a desired property for modern CALL and CAPT systems. MDD systems can focus on different levels of speech production. Phoneme-level assessment provides more detailed information than word- or sentence-level assessment and may help learners to correct phonetic errors. Goodness of Pronunciation (GOP) [1] is one of the most widely used methods for phoneme-level mispronunciation detection. GOP generates posterior-based confidence scores for each phoneme in the canonical phoneme sequence based on acoustic models usually trained for automatic speech recognition (ASR). Many works propose to increase the accuracy of GOP, e.g., weighted-GOP [2], lattice-based GOP [3], and Force-aligned GOP [4]. The GOP can also be extended by introducing phonological rules from L1 to detect typical pronunciation errors of L2 learners as done in [5][6]. Because the GOP is usually individually evaluated for each phoneme in the canonical sequence, the lack of context awareness is one of the issues related to the original GOP. In [7], context-aware GOP is proposed to address this issue.

More recently, foundation models trained on massive amounts of data have been used to improve MDD. Methods based on self-supervised learning (SSL), e.g., wav2vec2.0 [8] and HuBERT [9] have been used as feature extractors in [10]–[12] where the models were fine-tuned to predicting mispronunciations. However, fine-tuning to the MDD task requires mispronunciation annotations that are usually difficult to get by in sufficient quantities. Consequently, Liu et.al. [13] used a similar approach, but without fine-tuning the representations to the task. All these methods rely on forced-alignment of phonemes with speech. In [14] a pure end-to-end method for mispronunciation detection is proposed. However, this method requires alignment between the human annotated and the canonical phoneme sequence during training.

Another way of using foundation models for MDD is to fine-tune powerful end-to-end systems to perform phoneme recognition [15]–[17]. The phonetic transcription generated by the model is compared with the canonical transcription to detect discrepancies. This limits the methods to only producing a binary assessment. The main difference between [15] and [16], [17] is that, in the latter, context-awareness is introduced by considering the canonical transcription as an input to the system.

In this paper, we overcome the limitations of the aforementioned methods in a number of ways that can be summarized in the following contributions:

- We first show that SSL models fine-tuned with CTC loss can be used in the traditional alignment-based formulation of GOP. This not only provides a graded assessment, but allows us to fine-tune the models for phoneme recognition instead of the MDD task directly, reducing the amount of mispronunciation annotations required.
- We propose a new framework to compute GOP that is not dependent on phoneme level alignment. This simplifies calculations especially for CTC models, and eliminates the influence of possible misalignment between phonemes and speech.
- We use this framework to compute both uni-dimensional and multi-dimensional GOP features.
- We prove the effectiveness of our method with experiments on the CMU-kids and speechocean762 datasets containing speech from children and adults.
- We make all the code available to improve reproducibility of the results.

2. Background

Phoneme based pronunciation assessment systems traditionally rely on acoustic models trained for speech recognition to measure deviations from the canonical pronunciation. Given

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the sequence of feature vectors $O_1^T = \{o_1, \dots, o_T\}$ of length T and its corresponding canonical phoneme transcription $L_{\text{cano}} = \{l_1, \dots, l_{|L_{\text{cano}}|}\}$, the original definition of GOP [1] for phoneme $l_i \in L_{\text{cano}}$ is defined by the log posterior probability of that label:

$$\begin{aligned} \text{GOP-GMM}(l_i) &= \log(p(l_i|O_{t_1}^{t_2}))/|t_2 - t_1| \\ &= \log\left(\frac{p(O_{t_1}^{t_2}|l_i)P(l_i)}{\sum_{q \in Q} p(O_{t_1}^{t_2}|q)P(q)}\right)/|t_2 - t_1|, \end{aligned} \quad (1)$$

where $O_{t_1}^{t_2}$ are the feature frames corresponding to the canonical phoneme l_i . $P(q)$ is the prior probability for any phoneme q in the phoneme inventory Q and it is often neglected by convention. The term $p(O_{t_1}^{t_2}|l_i)$ was traditionally modeled by HMM-GMM-based acoustic models. The final score is then normalized by the length of the segment.

More recently, GMMs have been substituted by DNN-based ASR models, which are usually trained with discriminative loss. A typical definition of DNN-based GOP is simply the average log-posterior[18]:

$$\text{GOP-DNN}(l_i) = \sum_{t=t_1}^{t_2} \log(p(s_t|O_t))/|t_2 - t_1| \quad (2)$$

where $p(s_t|O_t)$ can be obtained from the final output layer for each input frame.

A natural progression would be to use more recently proposed end-to-end methods trained with CTC loss [19] for this task. However, CTC-based ASR systems are characterized by a special blank (ϕ) output symbol. This causes these methods to produce a “peaky” behaviour [19]–[21]: The blank symbol is active for most time steps, whereas the character/phoneme outputs only become active most of the time for one time step per segment. This is often regarded as an indication of overfitting and an obstacle to obtain reliable alignment information for MDD systems. These models have, therefore only been used in combination with a decoder that generates a phoneme sequence to be compared to the canonical transcription.

In both Eqs. 1 and 2, the segmentation information, or alignment (t_1, t_2) , which is derived from forced-alignment, has a vital influence for the computation of the GOP. However, there are several aspects that may degrade the quality of alignment. Firstly, even with perfect pronunciation, the boundary between subsequent phonemes can be arbitrary because of co-articulation in speech production [7][22]. Secondly, speech variability may cause alignment failure due to, e.g., speaker’s age, health conditions, or L1 influence for L2 learners, to name a few ([23]–[25]). Lastly, mispronunciations can cause alignment errors and larger errors may be expected in case of insertions or deletions (in phoneme error rate or PER terminology) of phonetic segments.

Another issue related to mispronunciations for GOP is due to its evaluation defined by Eqs. 1 or 2. It assumes that each phoneme in the canonical pronunciation may only be substituted by another sound if mispronounced. However, it does not account for deletion and insertion errors.

3. Methods

We propose a number of methods to estimate scalar and vector GOP features using models trained with CTC. The first method uses standard forced alignment, whereas the following ones are alignment free.

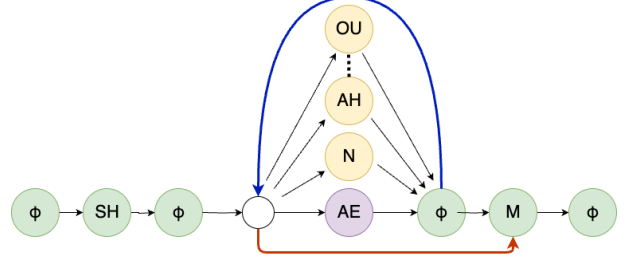


Figure 1: The graphs corresponding to the different GOP-CTC-alignment-free methods. The purple node (“AE”) is the target phoneme. Green nodes correspond to context in the canonical sequence “SH AE M”. Yellow nodes correspond to potential substitutions.

3.1. Forced-aligned CTC-Based scalar GOP

The first method we propose uses an ASR model fine-tuned with CTC for phoneme recognition to compute the GOP. We use a Viterbi decoder based on the same model’s logits to produce the alignment between frames and phonemes. As in GOP-DNN, we use Eq. 2 to estimate GOP, with the difference that only non-blank tokens contribute to the estimation, while the blank tokens are skipped. The hypothesis that motivates this method, and that we want to verify in this work, is that the peaky behavior of CTC-trained models may result in more reliable GOP values in case of mispronunciations, especially for deletions and insertions. We call this method GOP-CTC-align.

3.2. Alignment-Free CTC-Based scalar GOP

Next, we introduce a new alignment-free framework for computing GOP with CTC trained models with the goal of reducing the effect of alignment errors. The method takes as input a sequence of speech feature vectors O_1^T and the canonical phonetic transcription L_{cano} . Assuming that we are interested in assessing the pronunciation of a phoneme $l_i \in L_{\text{cano}}$, we can split the canonical transcription as

$$L_{\text{cano}} = \underbrace{\{l_1, \dots, l_{i-1}\}}_{\text{left context}}, \underbrace{l_i}_{\text{target phoneme}}, \underbrace{\{l_{i+1}, \dots, l_{|L_{\text{cano}}|}\}}_{\text{right context}}. \quad (3)$$

Computing the GOP as defined in Eq. 1 and Eq. 2, would require knowing the segmentation $[t_1, t_2]$, that is, when l_i starts and ends in the observation sequence O_1^T . In order to avoid this, we define the GOP as a ratio between the probability of the full canonical transcription $p(L_{\text{cano}}|O_1^T)$ and the probability of any transcriptions that share with L_{cano} the left and right context. We call $L_{\text{sdi}}(l_i)$ the set of transcriptions with these properties:

$$L_{\text{sdi}}(l_i) = \underbrace{\{l_1, \dots, l_{i-1}\}}_{\text{left context}}, \underbrace{*}_{\text{any phoneme sequence}}, \underbrace{\{l_{i+1}, \dots, l_{|L_{\text{cano}}|}\}}_{\text{right context}} \quad (4)$$

where “sdi” signifies that l_i can be either kept, deleted, substituted or there may be any number of insertions between the left and right context.

The corresponding definition for GOP is:

$$\text{GOP-CTC-AF-SDI}(l_i) = \log\left(\frac{p(L_{\text{cano}}|O_1^T)}{p(L_{\text{sdi}}(l_i)|O_1^T)}\right) \quad (5)$$

The rationale behind this definition is that, allowing any paths between the left and the right context corresponds to adding

a number of contributions close to 1 to the posterior. Consequently, the denominator in Eq. 5 acts approximately as a normalizing factor that tends to eliminate the effect of the left and right context from the numerator, leaving only the contribution from the target phoneme.

The numerator and the denominator in Eq. 5 can be efficiently computed by the forward algorithm using the graph shown in Figure 1. The numerator is computed restricting the graph to the horizontal path, including the green and purple nodes that correspond to the canonical transcription L_{cano} . For the denominator, we distinguish between three different variants of this method: In GOP-CTC-AF-SDI, the denominator in Eq. 5 is computed using the full graph, and all kind of errors (substitution, deletion and insertions) are considered. In GOP-CTC-AF-SD, the blue arrow is removed so that only a substitution or deletion is allowed. Finally, for GOP-CTC-AF-S, both red and blue arrows are removed so that only a substitution is allowed.

It is worth noting that the traditional definition of GOP only allows substitutions of the target phoneme. It is also worth to stress that the calculations in Eq. 5 are independent of the starting and ending times $[t_1, t_2]$ for label l_i or, equivalently, they marginalize out the value of t_1 and t_2 when computing the probabilities.

3.3. Alignment-Free CTC-based GOP feature vectors

The third method we propose is inspired by the GOP-based features introduced in [2], [26], [27] that we denote GOP-feature-TDNN in this paper.

Instead of computing a single GOP value using the graph in Figure 1 as in Section 3.2, we enumerate all possible transcription sequences from a subset of $L_{\text{sdi}}(l_i)$ where we only allow substitutions or deletion of l_i . We call this set of transcriptions $L_{\text{sd}}(l_i)$. We then define the log posterior probability (LPP) and a vector of log posterior ratios (**LPR**) for phoneme l_i as:

$$\begin{aligned} \text{LPP} &= \log p(L_{\text{cano}} | O_1^T) \\ \mathbf{LPR}(i) &= \left\{ \log \left(\frac{p(L_{\text{cano}} | O_1^T)}{p(L | O_1^T)} \right) \text{ with } L \in L_{\text{sd}}(l_i) \right\} \end{aligned} \quad (6)$$

The features of the i th phoneme l_i is then defined to be:

$$\text{GOP-feature-CTC-AF}(i) = \{\text{LPP}, \mathbf{LPR}(i)\} \quad (7)$$

For example, if we have 39 phonemes in the phoneme inventory, then there are 40 dimensions in **LPR** corresponding to the canonical phoneme, 38 possible substitutions errors and one deletion error. Adding the LPP, the total length of the feature vector becomes 41:

$$|\text{GOP-feature-CTC-AF}(i)| = |\text{LPP}| + |\mathbf{LPR}(i)| = 1 + 40 = 41.$$

Similarly to the method proposed in Section 3.2 and differently to the GOP-based features in [26], this feature vector does not rely on any specific alignment.

4. Experiments

We perform experiments on a well-known dataset of child speech (CMU-kids) and on a dataset specifically designed for pronunciation assessment (Speechocean762).

4.1. CMU-kids Data

The CMU-kids [28] comprises 9.1 hours of speech from children in the range from six to eleven years old. There are in total

5180 utterances spoken by 24 male and 65 female speakers. By comparing the phonemic transcriptions provided by the CMU-kids to a dictionary with canonical pronunciations, we are able to annotate each phoneme as 1 (missing or mispronounced) or 0 (correctly pronounced). We refer to this kind of errors as real errors. For the utterances without errors, following the work of [29], we synthesize mispronunciations systematically to increase the number of mispronounced samples. We call these simulated errors.

4.2. CMU-kids Evaluation

We only use the scalar GOP features for this data. This and the fact that the task is binary, allows us to use a threshold-free performance assessment criterion, namely the AUC-ROC. As the baseline method for CMU-kids, the best method "DNN-tri-frame" from [29] is used.

4.3. Speechocean762

The Speechocean762 [30] is a public accessible dataset designed for pronunciation assessment. The dataset comprises 5,000 utterances spoken by 250 non-native gender balanced speakers with an equal ratio of children and adults. Speechocean762 provides rich label information annotated by humans. Among those, we focus on the phoneme-level annotations. Differently from the CMU-kids binary assessment task, Speechocean762 defines three grades: 0 (mispronounced), 1 (strongly accented) and 2 (correctly spoken).

4.4. Speechocean762 Evaluation

In order to compare with published results on this task, we use GOP feature vectors from Section 3.3 in addition to the scalar GOP methods used for CMU-kids. We use Speechocean762 recommended methods for finding decision thresholds in the models. For scalar GOP features polynomial regression of order two is used. For the GOP vector features, we use support vector regression (SVR) to compare our method (GOP-feature-CTC-AF) to the vector baseline (GOP-feature-TDNN). Finally, we also test our GOP-feature-CTC-AF on the more recently proposed GOPT model [31] that predicts the grades end-to-end. This model can be trained both for only phoneme assessment or multi-task (phoneme, word, sentence). When we tried to train this model with the available code in the multi-task mode, however, the results for the phoneme assessment have a very high variance. We therefore only compare our model to the phoneme-only version of GOPT.

For evaluation, following common practice for this task, we report the Pearson Correlation Coefficient (PCC) and its confidence intervals.

4.5. Implementation of GOP methods

For all the baseline methods GOP-DNN, GOP-TDNN and GOP-feature-TDNN, we follow the same Kaldi [32] recipes as their original papers for training the models and generating the GOP. All of the non-baseline methods are based on wav2vec2-large-xlsr-53 [33]. This model is pre-trained in an unsupervised way with multi-language resources which is often regarded to be beneficial for the MDD task. The pre-trained model's output layer is then replaced by a fully connected layer for fine-tuning of a phoneme-recognizer. The target is one of the 39 English phonemes as defined in the Kaldi's recipe for Librispeech [34]. The model is fine-tuned either by using the Cross-Entropy-Loss (for GOP-CE) or CTC (for GOP-CTC-X),

Table 1: AUC values for CMU-kids, both simulated error and real error

AUC value for CMU-kids		
Features	Simulated error	Real error
GOP-DNN[29]	0.824	0.796
GOP-CE	0.967	0.851
GOP-CTC-align	0.988	0.905
GOP-CTC-AF-S	0.989	0.891
GOP-CTC-AF-SD	0.986	0.915
GOP-CTC-AF-SDI	0.938	0.859

with the same native English dataset “train-100-clean” from the LibriSpeech. For fine-tuning with the Cross-Entropy-Loss, the frame-level alignment information is obtained by a context-independent GMM phone model trained with the standard Kaldi recipe “train_mono.sh” on the “train-100-clean”. No alignment is required for the CTC-based fine-tuning.

The models are then used to compute all the different variants of GOP². For GOP-CE we used Eq. 2, whereas all the CTC-based definitions are given in Section 3. The forward algorithm used in computing the denominator in Eq. 5 is a modified version of the Stanford’s CTC loss [35].

5. Results

5.1. CMU-Kids

The experimental results for CMU-kids are shown in Table 1. We first observe that the GOP-CE largely outperforms the baseline method GOP-DNN, even though using the same training data (train-100-clean), loss function and alignments from the same model. This can be explained by the power of the multi-language pre-training step and the relative complex transformer network of wav2vec2.0 compared to a simple DNN model.

Secondly, we notice an improvement using GOP-CTC-align compared to GOP-CE. We hypothesise that the CTC’s “peaky” behaviour could alleviate the effect of alignment errors and mispronunciations mentioned in Section 2. The reason could be that in the CTC logits, the non-blank token activations become more temporally constrained, being ‘isolated’ by the blank tokens, which results in less interference from the neighbouring phonemes in terms of segmentation. This is also confirmed by the results on real errors, where GOP-CTC-align outperforms GOP-CTC-AF-S that only considers one type of mispronunciations (substitutions).

The two alignment-free methods outperform any other alignment-based methods: GOP-CTC-AF-SD performs best on the real errors while GOP-CTC-AF-S works best for simulated errors. The reason can be explained by the way of the error simulation that is followed in [29]. The errors simulated by replacing canonical phonemes systematically can only mimic substitution errors and that is exactly the type of errors GOP-CTC-AF-S captures. In reality, deletions and insertions might also happen. However, GOP-CTC-AF-SDI, which is supposed to address all possible mispronunciations, does not meet our expectation. We interpret it as the limitation of the metrics that we are using here. Because AUC-ROC and PCC are both evaluated with respect to the phonemes in the canonical phoneme sequence which limits the type of errors only to substitution and deletion. Therefore, the metric is not suitable to assess cases with multiple insertion errors shadowing the advantage of using GOP-CTC-AF-SDI.

²<https://github.com/frank613/CTC-based-GOP.git>

Table 2: Results for Speechocean762, reported with Pearson correlation coefficients

Speechocean762		
Features	MDD model	PCC
GOP-TDNN[30]	poly. reg.	0.361±0.008
GOP-CE	poly. reg.	0.378±0.007
GOP-CTC-align	poly. reg.	0.389±0.007
GOP-CTC-AF-S	poly. reg.	0.410±0.007
GOP-CTC-AF-SD	poly. reg.	0.433±0.007
GOP-CTC-AF-SDI	poly. reg.	0.373±0.007
GOP-feature-TDNN[30]	sup. vec. reg.	0.441±0.007
GOP-feature-CTC-AF	sup. vec. reg.	0.568±0.006
GOP-feature-TDNN[31]	GOPT	0.605±0.002
GOP-feature-CTC-AF	GOPT	0.618±0.003

5.2. Speechocean762

For Speechocean762, as seen in Table 2, the trend is consistent with the observation from the CMU-kids experiments in that GOP-CTC-AF-SD still performs best as a scalar feature in terms of PCC. It can be observed from the table that our best result of GOP-CTC-AF-SD is significantly better than any other scalar GOP methods in terms of confidence intervals. With the vector GOP features (GOP-feature-CTC-AF), we obtain 29.02% relative improvement on PCC over the baseline GOP-TDNN-feature using support vector regression and a 2.14% relative improvement when using the GOPT model. All the differences are statistically significant. To the best of our knowledge, the highest PCC reported for the phoneme level on Speechocean762 is 0.69 from [36]. However, their method is not comparable with ours because they use more contextual information as input and more complicated model for multi-task training which is besides the focus of this paper.

6. Conclusions

In this work, we propose the use of CTC-based models for phoneme level MDD. We define a number of methods for estimating scalar and vector GOP features. We explore both methods that require explicit segmentation of speech into phonemes, and a framework to compute GOP features that is segmentation (or alignment) free. This framework also accounts for insertions and deletions as well as substitutions as pronunciation errors.

We show with experiments on the CMU-kids and the Speechocean762 datasets that our methods are competitive with respect to the state-of-the-art. On CMU-kids, the alignment-free scalar GOP feature that only considers substitutions obtains the best results on simulated errors. On the other hand, to tackle real errors the best results are obtained with the GOP method that also allows deletions. On the Speechocean762 data, we perform experiments with both scalar and vector GOP features. For the vector features, we perform pronunciation assessment with both support vector regression and GOPT. In all cases, our alignment-free GOP features based on substitutions and deletions obtain the best results. This confirms that accounting for deletion errors is important in pronunciation assessment, something that only an alignment free model can achieve.

Finally, we believe that the proposed framework can be extended to using other end-to-end models, allowing for even further performance improvements.

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